POST-QUANTUM CRYPTOGRAPHY (PQC) PLATFORM

Complete Technical Presentation Master Notes & Speech Guide (Full Tanglish)

1. Problem Statement & Why PQC? (The Core Threat)

Slide Points (PPT Content):

- **The Quantum Threat:** Classical Cryptosystems (RSA, ECDH, ECDSA) rely on Prime Factorization & Discrete Logarithms. Quantum computers executing **Shor's Algorithm** break them instantly in polynomial time.
- **Grover's Algorithm:** Halves symmetric key bit-security (reduces 256-bit to 128-bit brute-force strength).
- **HNDL Attack (Harvest Now, Decrypt Later):** Adversaries store today's encrypted wiretap traffic to decrypt once Cryptographically Relevant Quantum Computers (CRQCs) arrive.
- **Solution:** Lattice-based Key Encapsulation (FIPS 203), Digital Signatures (FIPS 204), & Hash-based signatures (FIPS 205).

■■ ***Presentation-la Neenga Pesa Vendiyadhu (Tanglish Speech):***

*• 'Namma daily use panra WhatsApp, Banking, HTTPS ellame RSA & ECC encryption-la thaan irukku. Aana Quantum Computer vandha, **Shor\'s Algorithm** use panni indha encryption-ai romba easy-ah break pannidum.'*

*• 'Romba per ninaikiranga Quantum computer varumbodhu pathukalam-nu. Aana ippove hackers **HNDL (Harvest Now, Decrypt Later)** panranga. Adhavadhu innaiki pora encrypted traffic-ai store panni vechitu, future-la quantum computer vandhadhum decrypt panni data-va thiruduvaanga. Idhukku ore solution thaan namma **Post-Quantum Cryptography (PQC) Platform**.'*

2. Cryptographic Algorithms Inventory (NIST Standards)

Algorithm	Category & Standard	Key Size / Overhead	Function & Operational Role in System
ML-KEM (Kyber-768/1024)	Post-Quantum KEM (NIST FIPS 203)	Pub: 800 - 1568 B Cipher: 768 - 1568 B	Module-LWE lattice key encapsulation for quantum-safe shared secrets in 2-party chat.
ML-DSA (Dilithium-65)	Post-Quantum Digital Sig (NIST FIPS 204)	Pub: 1312 - 2592 B Sig: 2420 - 4595 B	Signs handshakes & packets to stop MITM attacks and guarantee non-repudiation.
SLH-DSA (SPHINCS+)	Stateless Hash-Based Sig (NIST FIPS 205)	Pub: 32 - 64 B Sig: 7.8 - 49 KB	Conservative fallback identity signing. Relies purely on hash security (SHA-2/SHAKE).
X25519 (ECDH)	Classical Key Exchange (RFC 7748)	Pub: 32 B Secret: 32 B	Paired with ML-KEM in **Hybrid Mode** for dual defense-in-depth security.
AES-256-GCM	Authenticated Bulk Cipher (NIST SP 800-38D)	Key: 32 B (256-bit) Nonce: 12 B, Tag: 16 B	Encrypts message bodies & files with Galois/Counter hardware-accelerated integrity.
Ascon-128a	NIST Lightweight Crypto (NIST LWC)	Key: 16 B (128-bit) Nonce/Tag: 16 B	Lightweight authenticated encryption engine for IoT edge devices & microcontrollers.
HKDF-SHA256	Key Derivation (RFC 5869)	Input: Variable Output: 32 B	Extracts & expands shared secrets (X25519 + ML-KEM) into symmetric session keys.

■■ ***Presentation-la Neenga Pesa Vendiyadhu (Tanglish Speech):***

*• **'ML-KEM (Kyber):** Idhu Key exchange-ku use aagudhu. Multi-dimensional lattice math base panni quantum-safe shared secrets exchange panrom.'*

*• **'ML-DSA & SLH-DSA:** Idhu Digital Signatures. Identity verify panni Man-in-the-Middle (MITM) attack-ai total-ah thadukudhu.'*

*• **'Hybrid Mode:** ML-KEM kooda classical X25519-ai combine panrom. Future-la lattice math-la flaw vandhalum classical 128-bit security protect pannum. Both keys-ai HKDF-SHA256-la pottu 256-bit symmetric session key generate panrom.'*

3. End-to-End Cryptographic Workflow (5 Execution Phases)

- **Phase 1 (Initiation):** User A generates Ephemeral X25519 + ML-KEM-768 keypairs; sends *initiate_handshake* via WebSocket.
- **Phase 2 (Mutual Consent):** User B receives real-time modal with User A's verified cryptographic ID & mode. Must click 'Accept' (No unilateral key forging).
- **Phase 3 (Key Derivation & Sig Verify):** User B encapsulates secret via ML-KEM + X25519; signs payload with ML-DSA-65. User A verifies signature before decapsulation. HKDF derives symmetric session key.
- **Phase 4 (Message Encryption):** Client increments monotonic sequence number; generates 96-bit random IV; encrypts via AES-256-GCM. AAD set to *'{mode}-{sequence_number}'*.

• **Phase 5 (Reception & Anti-Replay):** Receiver checks sequence number against replay cache; verifies signature; GCM Auth Tag verifies 0-bit tampering.

■■ ***Presentation-la Neenga Pesa Vendiyadhu (Tanglish Speech):***
• *'**Phase 1 & 2:** User A handshake start pannumbodhu temporary keys create aagi User B-kku pogum. User B consent kudutha mattum thaan handshake continue aagum.'*
• *'**Phase 3:** User B ML-DSA signature pottu anupuvaaru. Signature verify aana mattum thaan decapsulate aagi HKDF moolama session key generate aagum.'*
• *'**Phase 4 & 5:** Message anupumbodhu Sequence Number (1, 2, 3...) & AAD bind panni AES-GCM-la encrypt panrom. Receiver-kku pogumbodhu duplicate sequence number vandha **Replay Attack**-nu alert adikkum. Single bit data maathinaalum GCM Tag decapsulation fail aagi block pannidum.'*

4. Advantages (Pros) vs Technical Trade-offs (Cons)

Advantages (Pros)	Trade-offs & Constraints (Cons)
• **Quantum-Immunity:** Full protection against Shor's & Grover's attacks. • **Hybrid Defense-in-Depth:** Dual X25519 + ML-KEM ensures classical + post-quantum safety. • **NIST Standardization:** Aligns with NIST FIPS 203, 204, 205 (Aug 2024). • **Zero-Trust Identity:** Every handshake & message signed with ML-DSA/SLH-DSA. • **Anti-Tamper & Anti-Replay:** AES-GCM Auth tags + monotonic sequences. • **Interactive Attack Lab:** Simulates live MITM, replay, & key mismatch attacks.	• **Key & Ciphertext Size:** ML-KEM-768 public key is 1,184 B vs Classical 32 B. • **Bandwidth Consumption:** High-frequency group chats require more bandwidth. • **CPU / Memory Overhead:** Embedded/IoT devices need SIMD (AVX2/NEON) vector optimization for sub-millisecond speeds. • **SLH-DSA Signature Size:** Hash-based signatures range from 7.8 KB to 49 KB (heavy for live chat).

■■ ***Presentation-la Neenga Pesa Vendiyadhu (Tanglish Speech):***
• *'Pros paatha, NIST 2024 standards full-ah follow panrom, Zero-trust architecture, attack lab kooda integrate aagi irukku.'*
• *'Cons/Trade-off-nu paatha, Classical-ai vida PQC key & signature sizes perusu (Bytes-la irundhu KB level). Idhanaala bandwidth & CPU resource konjam extra thevaipadum, adhukku AVX2/NEON optimizations thevaipadum.'*

5. Industry Comparison Matrix (Competitive Analysis)

Platform	Key Exchange	Digital Signatures	Symmetric Cipher	PQC Status
Our PQC Platform	Hybrid X25519 + ML-KEM	ML-DSA-65 & SLH-DSA	AES-256-GCM / Ascon-128a	**FULLY RESISTANT (FIPS 203/204/205)**
Signal	PQXDH (ML-KEM-768)	Ed25519 (Classical)	AES-256-CBC / HMAC	PARTIALLY RESISTANT (Signatures classical)
Apple iMessage (PQ3)	Kyber-768 + P-256 Ratchet	ECDSA (Classical)	AES-256-GCM	PARTIALLY RESISTANT (Signatures classical)
WhatsApp	Curve25519 (Classical)	Ed25519 (Classical)	AES-256-GCM	VULNERABLE (No PQC support)
Telegram	MTProto 2.0 (Classical DH)	RSA-2048 (Classical)	AES-256-IGE	CRITICAL VULNERABILITY (Broken by Shor's algo)

■■ ***Presentation-la Neenga Pesa Vendiyadhu (Tanglish Speech):***
• *'**WhatsApp & Telegram** innum classical RSA/ECC-la irukku, quantum attacks-ku 100% vulnerable.'*
• *'**Signal & Apple PQ3** Key exchange-ku PQC kondu vandhutaanga, aana avangala Signature innum classical Ed25519/ECDSA thaan.'*
• *'**Namma Platform:** Key Exchange, Digital Signatures, Symmetric ciphers ellathulayume **100% Full Post-Quantum Resistant** architecture implement pannirukom.'*

6. Commercial Cloud Deployment & Cost Breakdown

Infrastructure Component	Recommended Cloud Tier	Monthly (USD)	Annual (USD)
App & WebSocket Cluster	3x AWS c6i.xlarge (Compute Optimized) behind ALB	$260 / mo	$3,120 / yr
Managed Database Cluster	AWS RDS PostgreSQL (Multi-AZ, db.r6g.large)	$220 / mo	$2,640 / yr
Hardware Security Module	AWS CloudHSM / Azure Managed HSM for PKI Root	$180 / mo	$2,160 / yr

Encrypted Object Storage	AWS S3 (SSE-KMS, 5 TB storage + Data Egress)	$125 / mo	$1,500 / yr
TURN / STUN Relays (VoIP)	2x Coturn edge nodes (NAT traversal & low-latency)	$60 / mo	$720 / yr
CDN & DDoS Defense Shield	Cloudflare Business/Enterprise with WAF & Rate Limit	$200 / mo	$2,400 / yr
Cryptographic Audit (Year 1)	External audit by NCC Group / Trail of Bits (One-Time)	N/A	$12,000 (Yr 1)
TOTAL ESTIMATED BUDGET	**Enterprise Scale (10k - 50k Concurrent Users)**	**~$1,045 / mo**	**~$24,540 (Yr 1)**

■■ ***Presentation-la Neenga Pesa Vendiyadhu (Tanglish Speech):***
• 'Commercial production deployment cost romba viable-ah irukku. 10,000 to 50,000 active users support panna masathukku approx ***$1,045 (~Rs. 87,000)*** *thaan aagudhu. Idhula High compute AWS servers, CloudHSM for Root keys, Multi-AZ database and Cloudflare DDoS protection ellame include aagirukku.'*

7. Practical Real-World Applications & Industry Sectors

• **Defense & Military Intelligence (NATO / DoD Compliance):** Military data secrecy 30 to 50 years thevaipadum. Innaiki intercept panna classified traffic-ai future quantum decryptions-la irundhu protect pannum.

• **Financial Networks & Interbank Communications (SWIFT / Fedwire):** Trillions of dollars wire transfers & payments nadakkum financial backbones-la quantum replay & signature spoofing thadukudhu.

• **Healthcare & Genomic Data Repositories (HIPAA / GDPR):** Patient DNA sequencing & clinical medical records life-long sensitive. HNDL attacks-la irundhu permanent secrecy tharum.

• **Critical Infrastructure & Industrial IoT (SCADA / Power Grids):** Ascon-128a & lightweight ML-KEM-512 use panni power stations, water grids & smart telemetry-ai state-sponsored cyber attacks-la irundhu protect pannum.

• **Executive Corporate Communications:** M&A; deals, trade secrets, patents & intellectual property transfer safe-ah nadakka udhavum.

■■ ***Presentation-la Neenga Pesa Vendiyadhu (Tanglish Summary Pitch):***
• 'To conclude, Post-Quantum Cryptography oru future option illa, innaikiye thevaiyana immediate necessity. Defense, Banking, Healthcare, SCADA grids and Enterprise communications-kku namma platform 100% NIST FIPS 203, 204, 205 compliant, cost-effective and fully quantum-proof security-ai tharudhu. Thank you!'